

Generation Interconnection Request System Impact Study Report

For Queue Project Q20-1597

Gadsden 138 kV
November 29, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1597. The Interconnection Customer proposes a 142.8 MW wind facility interconnecting at the Gadsden 138 kV substation in the Pennsylvania area. The request was received on October 14, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 574 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$8,750,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of SALMON CREEK to DEPOE BAY 138 kV, the FLORENCE MILL to GADSDEN 138 kV loads to 149.3% of its 192 MVA rating. Project DFAX: 100.0%.

The DEPOE BAY to SALMON CRK 138 kV L is overloaded to 132.9% of its 216 MVA emergency rating for the loss of the FLORENCE MILL to GADSDEN SUBSTATION 138 kV. The project contributes 100.0% DFAX (142.8 MW) to this overload.

For the loss of BLACKROCK to SHAWVILLE STA SUBSTATION 345 kV, the SHAWVILLE STA 345/138kV Auto loads to 112.8% of its 326 MVA rating. Project DFAX: 23.0%.

For the loss of GADSDEN to FLORENCE MILL 138 kV, the TEMPLE SWITCHING SUBSTATION to LARNED 138 kV loads to 105.4% of its 192 MVA rating. Project DFAX: 56.5%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
FLORENCE MILL to GADSDEN 138 kV	Rebuild line with higher-capacity conductor	\$3,318,000	32% (\$1,056,000)
SALMON CREEK to DEPOE BAY SUBSTATION 138 kV line	Rebuild line with higher-capacity conductor	\$13,245,000	32% (\$4,215,000)
SHAWVILLE STA 345/138kV Auto	Replace transformer with higher-rated unit	\$10,070,000	30% (\$3,065,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
LARNED to TEMPLE SWITCHING 138 kV Line	Reconductor line	\$1,302,000	32% (\$414,000)
	Total allocated to project		\$8,750,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.